COINCIDENT PEAK PREDICTION USING A FEED-FORWARD NEURAL NETWORK

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ABSTRACT

A significant portion of a business' annual electrical payments can be made up of coincident peak charges: a transmission surcharge for power consumed when the entire system is at peak demand. This charge occurs only a few times annually, but with per-MW prices orders of magnitudes higher than non-peak times. A business is incentivized to reduce its power consumption, but accurately predicting the timing of peak demand charges is nontrivial. In this paper we present a decision framework based on predicting the day-ahead likelihood of peak demand charges. We train a feed-forward neural network to estimate the probability of system demand peaks and show it outperforms conventional forecasting methods using historical load. Using ERCOT demand and weather data from 2010-2017, we show the effectiveness of our framework.

1. INTRODUCTION

Coincident peak (CP) charges are a pricing mechanism used by some electrical utilities and system operators to incentivize customers to reduce consumption during system demand peaks [1, 2]. Since a CP charge is on the order of 100 times of regular electricity prices [3, 4], it accounts for a significant portion (upwards of 20% [4]) of a business' total annual electrical costs. The Public Utility Commission of Texas (PUCT) for example, levies coincident peak charges annually through a voluntary program during each of the summer months to set transmission fees for participating consumers [3]. In exchange for participation, consumers benefit from discounted electrical rates at all other times. Other CP charging schemes—such as those where utilities announce an upcoming peak ahead of time—are also used, however we are interested in CP charging schemes where the peak is observed *after the fact*.

For a business to reap such CP charge savings, the challenge lies in predicting when a peak will occur. A business would need to predict the *system-wide demand*, which is not normally done at the customer level. Consequently a system operator typically provides warning signals such as the expected system peak demand or when a peak is expected to occur. Current literature on CP charges focuses on how business can utilize these signals.

These warning signals sent by system operators, however, are not always reliable. For example, ERCOT estimates monthly system peak demands via quantile regression [5] to provide a basis for warning signals. But in 2017, the actual ERCOT peak loads were roughly 4,000 MW less than predicted value, leading to business under-curtailing their load [5]. To avoid these type of errors, some other operators provide more conservative signals. For example, Fort Collins PUD typically sends out warning signals for 10 days of the month, usually with only 5 to 10 minutes of lead time. Although these signals are useful to make load curtailment decisions [4], their frequency, high false alarm rates, and short lead time is undesirable and may lead to user fatigue.

Both works [6] and [1] showed that businesses likely curtail their demand if enough time is given before the occurs. In this paper we present a framework where a business can use open source data to make curtailment decisions up to 24 hours in advance. Instead of trying to provide a binary forecast of whether the system peak would occur at a given time, we use a "softer" decision rule: we estimate the cumulative distribution function (CDF) of the load. This information provides a likelihood of the peak occurring and a business can adjust its demand according to this likelihood. Specifically, we train a neural network using open source historical load values, timing information (e.g., day of the week) and weather data. We show that this neural network out performs conventional CP prediction methods based on load forecasting.

The rest of the paper is organized as follows: Section 2 defines the technical problems solved, Section 3 provides the solution framework, Section 4 presents simulation results using ERCOT data and Section 5 concludes the paper.

2. PROBLEM STATEMENT

An individual consumer subject to CP charges is incentivized to reduce consumption during a CP. We suppose a consumer exhibits a generic utility g as a function of hourly power consumed at time t, p_t, over the course of a year's hours τ. We assume g is concave, monotonically increasing, and continuous [7] incorporating, for example, hourly electricity rates. Let π_{cp} be the coincidental peak charge rate and S_t be the total *system* power demands, $\{S\} := \{S_1, \ldots, S_\tau\}$. Let $r_n(\{S\})$ be a ranking function which yields the set of indecies t of the n largest values in $\{S\}$. For example, $r_1(\{S\})$ is the time index of the maximum value in $\{S\}$.

To maximize utility at time t subject to one CP charge over time horizon τ (an additive cost to utility), a business solves the following optimization program:

$$\begin{aligned} \underset{p_t}{\text{maximize}} \quad & g(p_t) - \pi_{cp} p_t \cdot \mathbf{1}\left[t \in r_1(\{S\})\right] \\ \text{subject to} \quad & p_t \le p_{\max} \\ & p_t \ge 0 \end{aligned} \tag{P1}$$

where $p_{\max}$ is the maximum power consumption. Note the indicator function in the CP charge term is equal to 1 *only* during a system peak, i.e. $S_t \ge S_s, \forall s \in \{0, \ldots, \tau\}$.

The challenge in solving (P1) is that the indicator function $\mathbf{1}\left[t = r_1(\{S\})\right]$ is non-causal, since we do not know whether a particular hour is the peak demand until all future demand in the period τ is observed. Hence, we need to make a curtailment decision while treating the indicator function as a random variable.

2.1. Equivalent formulation

A natural solution to (P1) is to directly predict whether the peak demand occurs at a particular time or not. However, this prediction is extremely difficult since it is essentially predicting a very rare event, with a high cost if the event is missed [8, 9]. Therefore we replace the indicator function with a probabilistic expression. Here we use the empirical cumulative distribution function (CDF), defined as:

$$F_\tau(x) = \frac{1}{\tau} \sum_{t=1}^{\tau} \mathbb{1}[S_t \le x]. \tag{1}$$

Using F, we can write (P1) in an equivalent form:

$$\begin{aligned} \underset{p_t}{\text{maximize}} \quad & g(p_t) - \pi_{cp} p_t \cdot \mathbf{1}\left[F_\tau(S_t) \ge 1\right] && \text{(2a)} \\ \text{subject to} \quad & p_t \le p_{\max} && \text{(2b)} \\ & p_t \ge 0 && \text{(2c)} \end{aligned}$$

where the indicator function only evaluates to be one when S_t is a system peak. Again, if all $\{S\}_{t=1}^{\tau}$ are known, then $F_\tau(x)$ can be determined directly, however, we are interested in the case where only $\{S\}_{t=1}^{m}$ for $(m < \tau)$ have been observed and a predictor $\hat{F}_\tau(x)$ is required. To avoid overloading notation, we drop the τ subscript with the understanding the empirical CDF F is over a fixed time horizon. The next section shows how (2) can be modified to use an estimated CDF.

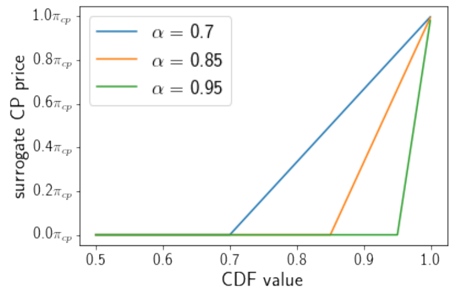


Fig. 1: Hedged CP cost as a function of the predicted CDF value for various values of α.

3. SOLUTION FRAMEWORK

3.1. Surrogate Problem

In practice—when the future information is not known—instead of comparing $F(S_t)$ with 1, we can relax the curtailment threshold. Let $\hat{F}$ be some estimation of the true CDF F, then we write the following surrogate problem for (2):

$$\begin{aligned} \underset{p_t}{\text{maximize}} \quad & g(p_t) - \pi_{cp} p_t \cdot \mathbf{1}\left[\hat{F}(S_t) \ge \alpha\right] && \text{(3a)} \\ \text{subject to} \quad & p_t \le p_{\max} && \text{(3b)} \\ & p_t \ge 0. && \text{(3c)} \end{aligned}$$

Here α "softens" the curtailment decision based on the predicted value of the CDF at time t. If $\hat{F}(S_t) \ge \alpha$, we then curtail the optimal planned power from $p_{\max}$ to p^* by interpolation over the interval $[\alpha, 1]$. The parameter α can then be tuned to suit the peak predictor $\hat{F}$. We now propose the following optimization program:

$$\begin{aligned} \underset{p}{\text{maximize}} \quad & g(p_t) - \pi_{cp} p_t \cdot \max\left\{0, \frac{F(S_t) - \alpha}{1 - \alpha}\right\} \\ \text{subject to} \quad & p \le p_{\max} \\ & p \ge 0 \end{aligned} \tag{P2}$$

The max term in the optimization problem acts as a hinge function (illustrated in Fig. 1) that activates once the value of predictor $\hat{F}(S_t)$ is large enough. Given a predictor with perfect knowledge and letting $\alpha = \frac{1}{8760}$, this surrogate optimization problem reduces to (P1) for each hour t.

Given our problem setup, our CP prediction efforts diverge from ERCOT market operations in one significant way; in order to highlight the difficulty of predicting relatively rare events, our goal in this work will be to predict the top 10 *annual* CP's, each at least 24 hours in advance, regardless of the month of occurrence. Our hypothetical business will be charged individually for each peak. For a thorough review of PUTC CP billing and amortization schedules, we refer the reader to [6].

In the rest of this section we construct a CDF value predictor based on the empirical CDF of historical system demand data: hourly ERCOT system demand data from 2010-2017 [10]. Our main predictor, a feed-forward neural network (NN), utilizes both historical system demand data and weather forecast data. In our application scenario, a CDF predictor $\hat{F}$ accepts a system power S_t along with various engineered features as a function of S_t and t to predict the proceeding 24 hours of CDF values. All relevant code and data can be found on our GitHub page github.com/cpatdowling/peakload.

3.2. Historical average predictor

As a baseline prediction method we create a historical average model for ERCOT. For each year from 2010-2016, we calculate an average annual system demand series $\{\bar{S}\}$ where each hour t is calculate as the average demand across years:

$$\bar{S}_t = \frac{1}{7} \sum_{year=2010}^{2016} S_t^{(year)}. \tag{4}$$

Consistent with ERCOT load growth forecast methodology [11], all demand data is mean-shifted *a priori*. Then from $\{\bar{S}\}$ we calculate an empirical CDF $\hat{F}_{hist}(x)$. In order to use this model in a predictive context, $F(S_t^{(2017)})$ is predicted directly as $\hat{F}_{hist}(\bar{S}_t)$.

3.3. Neural network predictor

To make use of weather forecasts, we employ a deep feed-forward neural network to train an additional predictor $\hat{F}_{NN}$. Neural networks are non-linear function approximators recently demonstrating broad success in a number of application areas like image classification [12] and natural language processing [13]. In the context of electrical system load forecasting, neural networks have been employed extensively [14, 15, 16]. In this paper we make use of a feed-forward network with six hidden layers with sigmoid and tanh activations. Network parameters can be found on our GitHub page.

Input features include date and time information for the current hour. System demand data spans the prior three weeks for each of the nine ERCOT operational subregions, including first order statistics on each hour of each day within that period. Further, means and variances of system demands observed thus far through the billing cycle up to the current point are included. Open-source weather data features [17] include a 24 hour forecast of temperature, humidity, wind speed, wind bearing, visibility, pressure, and precipitation for each of the 19 largest cities in the ERCOT market region. A more detailed manifest of data input features can also be found on our GitHub page.

3.4. Loss function

To train a NN, a straightforward loss function to use is in this context is the mean absolute error or average L1 loss. When performing coincidental peak prediction, however, the average L1 loss forces the training of the predictor $\hat{F}$ to consider values from $[0, 1]$ with equal importance. This isn't necessary when we wish to constrain our decision making efforts to only the largest values of F for which a peak occurs. To rectify this we weight the loss function according to the true CDF value. We define the exponentially weighted average L1 loss (EW) as:

$$\mathcal{L}_\beta := \frac{1}{|\{P\}|} \sum_{x_t \in \{X\}} \left[\beta^{F(S_t)} |F(S_t) - \hat{F}(S_t)|\right]. \tag{5}$$

Increasing the parameter β sharpens the loss incurred by incorrect predictions at large, *true* CDF values.

3.5. Metrics

To evaluate the performance of our predictor $\hat{F}$ we define some metrics. Because we only care about peak values, our metrics focus on CDF values above a certain threshold c.

Definition 1 (Threshold Loss)**.** *Let $\mathcal{L}$ be the average L1 loss, the threshold loss $\mathcal{L}(c)$ is defined as*

$$\mathcal{L}(c) = \frac{1}{m} \sum_{S_t \in \{S\}} \begin{cases} |F(S_t) - \hat{F}(S_t)| & \text{if } F(S_t) \text{ or } \hat{F}(S_t) \geq c \\ 0 & \text{otherwise} \end{cases}$$

where m is the of number non-zeros, i.e.,

$$m = \sum_{\{S\}} \mathbf{1}[F(S_t) \text{ or } \hat{F}(S_t) \geq c].$$

Definition 2 (Binary precision)**.** *For some threshold c, the binary precision is defined as*

$$P = \frac{\sum_{\{S\}} \mathbf{1}[\hat{F}(S_t) > c \text{ and } F(S_t) > c]}{\sum_{\{S\}} \mathbf{1}[\hat{F}(S_t) > c]} \tag{6}$$

Definition 3 (Binary recall)**.** *For some threshold c, the binary recall is defined as*

$$R = \frac{\sum_{\{S\}} \mathbf{1}[\hat{F}(S_t) > c \text{ and } F(S_t) > c]}{\sum_{\{S\}} \mathbf{1}[F(S_t) > c]} \tag{7}$$

4. RESULTS

For NN training, 2010-2016 are used as training and validation data, with a random 90%/10% split respectively. 2017 is held out entirely as a test data set for both $\hat{F}_{hist}$ and $\hat{F}_{NN}$.

Training a predictor with EW loss over average L1 loss exhibits the expected consequence for higher CDF values, as

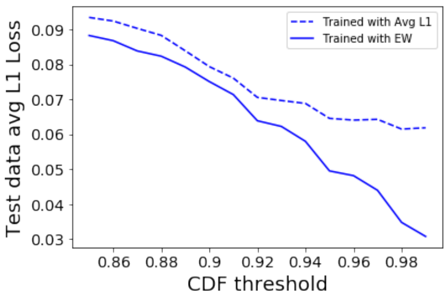
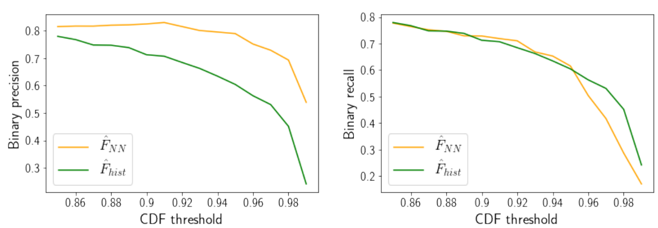


Fig. 2: Comparison of post-training average L1 loss after use of average L1 and EW loss during training with $\beta = 10$, identical network training parameters



(a) (b)

Fig. 3: Binary precision (3a) and binary recall (3b) for historical and NN CDF predictors

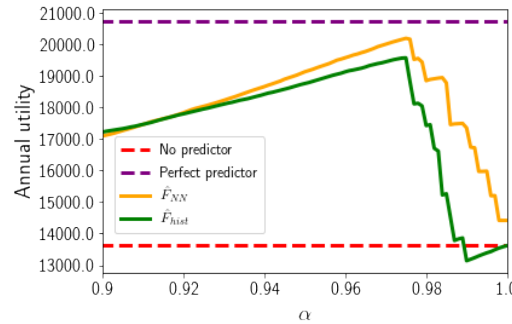


Fig. 4: Annual utility as a function of curtailment threshold alpha for historical and neural network predictors

is illustrated in Fig. 2. The threshold loss on test data outputs is lower for high CDF values for networks trained with EW loss. Since the goal is to accurately predict higher CDF values on test data, this approach provides a positive result.

4.1. Predictor performance

The NN, $\hat{F}_{NN}$, outperforms the historical average model $\hat{F}_{hist}$, for all CDF tresholds in binary precision, illustrated in Fig. 3a, correctly predicting CDF values over 0.98 at least 50% of the time. Both $\hat{F}_{NN}$ and $\hat{F}_{hist}$ have similar binary recalls, illustrated in Fig. 3b. To better understand a predictor's CP cost-savings performance, we turn to a hypothetical business utility model.

4.2. Utility model performance

Using program (P2) we develop a hypothetical business utility function of power to test the effectiveness of $\hat{F}_{NN}$ and the $\hat{F}_{hist}$, as measured against perfect knowledge and if a business takes no action. We presume our hypothetical business' utility function to maximize takes the form,

$$G(p_t) = log(1 + p_t) - \max\left\{0, \frac{F(P_t) - \alpha}{1 - \alpha}\right\} p_t \quad (8)$$

where $p_{\max} = 500$ MW. Operating during regular business hours (9 AM to 5 PM) on weekdays, this roughly equates to 10 utility per hour that isn't a CP. 10 CP hours are billed to the business over the course of the test year. The CP cost of 1 utility per MW is approximately 40% of the annual maximum utility of 21,000, representative of a more drastic CP cost scenario for a business.

Curtailment decisions for each regular business hour (9 AM to 5 PM) of each day are made at midnight based on 24-hour ahead predictions. Fig. 4 illustrates the performance of both $\hat{F}_{NN}$ and $\hat{F}_{hist}$ as a function of the curtailment threshold α. Utility for both predictors is maximized at $\alpha = 0.975$. The historical average model achieves approximately 94.4% of optimal annual utility, while $\hat{F}_{NN}$ model exceeds 97.4%. As α is increased to a tighter value, the purely time-of-year based predictor $\hat{F}_{hist}$ can perform suboptimally as compared to taking no action over the course of the year at all. Utility drops off as predicted CDF values are excluded by the curtailment threshold α.

5. CONCLUSION

In sum we considered how a business can curtail its demand based on CP charges. We formulated an optimization problem that overcomes the existing challenges of making a binary decision based on system peak predictions. Instead, we show how an estimate of the CDF can be used to achieve large CP charge savings with a neural network based predictor that outperforms existing historical-behavior based methods. The aim of future work is to explore several facets revolving primarily around developing 1) a more nuanced predictor and 2) testing it on a more realistic utility function.

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